

Document Summary TASK-3E54D148

Analysis & Generated Response

- **Title:** Sovereign AI Workbench for Confidential Industrial Work
- **Problem Statement ID:** SIH26117
- **Problem:** The current approach to confidential industrial work involves centralized cloud AI systems that are blocked by security policies and require manual intervention.
- **Solution:** We propose a sovereign AI workspace using open-weight multimodal LLMs, enabling on-premise agentic AI workflows for confidential data.
- **Theme:** Smart Automation
- **Theme:** The focus is on automating industrial processes through intelligent automation.
- **Team ID:** TEAM ADRESTIA
- **Team Members:**
 - Adrestia (Lead)
 - Adrestia (Member 1)
 - Adrestia (Member 2)
- **Problem Statement Title:** "Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work"
- **Team Name:** TEAM ADRESTIA

Detailed Solution:

INPUT

- PDF: Scan, Image, Drawing -> Code
- Local OCR + Vision tools (OCR AI and Vision)
- Private SOPs / Manuals -> Manual data management
- Open-source models from Qwen, Llama, Mistral, Gemma -> Model selection

UNDERSTANDING

Local OCR + Vision tools are used to extract text and images from PDFs. This allows for the creation of task plans using local OCR and vision capabilities.

RETRIEVE

Private SOPs / Manuals: Local models (OCR AI) can be retrieved without needing a central database, allowing for quick access to existing documentation.

PLAN

Agent creates a task plan based on the extracted data from PDFs. This plan is then executed by the model using local OCR and vision capabilities.

EXECUTE

Local models + tools are used to automate tasks such as document analysis, image processing, and data extraction.

- **Model Selection:** The agent selects the best local model for each task based on its performance in previous experiments.
- Example: Qwen is chosen for text-based tasks due to its ability to handle complex natural language processing (NLP) tasks efficiently.

VERIFY/ RERANK

Sandbox + evidence, select the best answer -> Evidence-driven decision-making process ensures that only the most accurate and relevant information is used in the model's output.

- **Evidence:** The agent uses a sandbox environment where it can test different models to identify the best one for each task

Grounding & Evidence Sources

- **final ppt sih adrestia.pdf** (Referenced): TITLE PAGE SMART INDIA HACKATHON 2026 • Problem Statement ID – SIH26117 • Problem Statement Title- "Sovereign On-Premise Agentic AI Workbench"